وزارة التعليم العالي
معهد العبور العالي
للإدارة والحاسبات ونظم المعلومات

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Obour Institutes

امتحان منتصف الفصل الدراسي الأول (لائحة 2018)

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Question [1]: Choose the correct answer:- (8 Marks)

1) The statement: $\sim(p \wedge \sim q)$ is equivalent to .....

a) $\sim p$	b) $(p \wedge \sim q)$	c) $(p \vee q)$	d) $p \rightarrow q$

2) The statement $p \rightarrow (p \vee q)$ is .....

a) $\sim p$	b) F	c) T	d) $\sim q$

3) The statement $(p \rightarrow q) \vee (p \rightarrow \sim q)$ is .....

a) T	b) F	c) q	d) $\sim p$

4) For two sets of elements A and B ,the Boolean algebra rule: $A + \bar{A}B = \cdots$

a) $\bar{A} + B$	b) $\bar{B} + A$	c) $A + B$	d) 1

5) If $A = \{0,1,2\}$, R is a relation on A where; $R = \{(0,2),(1,0),(1,2),(2,0)\}$, then $M_R =$ .....

a) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 \end{array}$ b) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 0 & 1 \end{array}$ c) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{array}$ d) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}$

6) If p, q, r are three logical statements where; p is true, q is true and r is false, then the statement $(\sim p \wedge \sim q) \vee r$ is .....

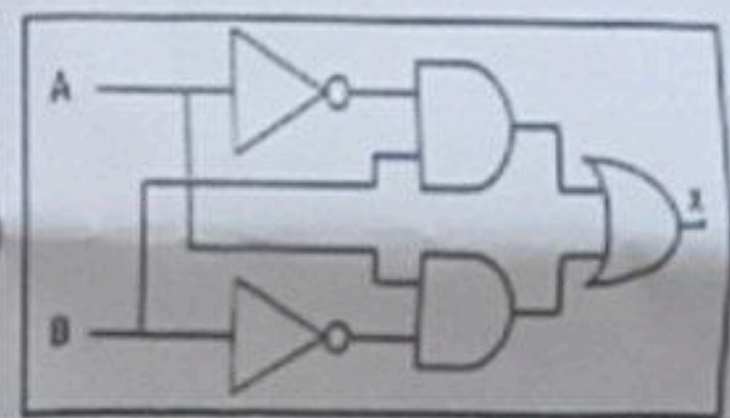
a) $\sim r$	b) $p \wedge q$	c) 1	d) 0

7) The statement $s \rightarrow r \equiv \ldots$

a) $s \vee r$	b) $s \wedge r$	c) $\sim s \vee r$	d) $r \rightarrow s$

8) In the following Schematic diagram:

$X = \cdots$



a) $\bar{A}B$	b) $\bar{A}B + A\bar{B}$	c) $\bar{B}A + B$	d) $A\bar{B}$

Question [2]: Answer in details the following questions:- (7 Marks)

i) If R is a relation defined on $A = \{a, b, c, d\}$ where;
$R = \{(a,a), (b,b), (c,c), (d,d), (c,a), (c,b), (d,a)\}$, represent R by Boolean matrix and discuss that R is an equivalence relation or not.

متماثلة

ii) Consider the argument: p ∴ q

Premise: If you bought bread, then you went to the store

Premise: You bought bread p

Conclusion: You went to the store q

Analyze it using truth tables by representing each of the premises symbolically and show is the conclusion is valid or not.

T F

Best Wishes ..... Dr/Saeed Hemeda

p	q	$p \to q$
1	1	1
1	0	0
1	0	1
0	1	1
0	0	1

$p \to (p \vee q)$

p	q	$p \vee q$	$p \to (p \vee q)$
1	1	1	1
1	0	1	1